

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10	U04						***** VALVES*****			B1	ALL BOLTS ARE METRIC DIA.
20	U04	VB2					***** BALL VALVES *****			B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
30											
30	U04		BALL	A01AS01	3	4	BALL 150# RF SS A182 F316 316SS TRIM FLOATING SOFT RB LO	0	0		
40	U04		BALL	A01AS03	6	6	BALL 150# RF SS A182 F316 316SS TRIM FLOATING SOFT RB GO	0	0	FS1	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.
50	U04		BALL	A01AS07	8	8	BALL 150# RF SS A182 F316 316SS TRIM TRUNNION SOFT RB GO	0	0		
60		VA5	BALL	B01AS01	0.5	3	BALL 150# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO	0	0		
70	U04	VA5	BALL	B01AS03	4	4	BALL 150# RF SS A182 F316 316SS TRIM FLOATING SOFT FB GO	0	0	GA6	SPECIAL GASKET ONLY FOR USE AS SPECIFIED IN ENG-200-117
80		VA5	BALL	B01AS07	6	16	BALL 150# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO	0	0		
90	U04		BALL	B01DS01	0.5	2	BALL 150# SCRD FNPT SS A182 F316 316SS TRIM FLOATING SOFT FB LO	0	0	NR1	SIZES SHOWN IN SIZE COLUMNS ARE THE SMALL END OF REDUCERS AND THE BRANCH SIZE OF OLETS AND TEES.
100	U04	FS1	BALL	B03AS01	0.75	2	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO	0	0		
110											
110	U04						***** BUTTERFLY VALVES *****			UD2	USE FOR VENTS AND DRAINS OR WHERE A THREADED OUTLET IS REQUIRED FOR INSTRUMENT CONNECTIONS.
120	U04		BUTTERFLY	T01AS61	10	16	BUTTERFLY 150# RF SS A351 CF8M 316SS TRIM TOV METAL DBL FLG SHORT PAT GO	0	0		
130											
130	U04						***** GATE VALVES *****			UG3	PTFE GASKET ONLY TO BE USED FOR INTERFACE WITH POLYETHYLENE (HDPE) UNDERGROUND PIPING
140											
140	U04		GATE	G01AS48	0.75	1.5	GATE 150# RF SS A182 F316 TRIM 16 WEDGE HW	0	0	UG4	BOLTS ONLY TO BE USED FOR INTERFACE WITH POLYETHYLENE (HDPE) UNDERGROUND PIPING
150	U04	FS1	GATE	G03AS48	0.75	0.75	GATE 300# RF SS A182 F316 TRIM 16 WEDGE HW	0	0		
160		FS1	GATE	G03AS41	2	16	GATE 300# RF SS A351 CF8M TRIM 16 WEDGE HW	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
170											
170	U04						***** GLOBE VALVES *****			VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
180											
180	U04		GLOBE	L01AS48	0.75	1.5	GLOBE 150# RF SS A182 F316 TRIM 16 OS&Y HW	0	0		
190	U04		GLOBE	L01AS41	2	6	GLOBE 150# RF SS A351 CF8M TRIM 16 OS&Y HW	0	0	VC1	SPRING PISTON CHECK VALVES ARE PREFERRED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
200	U04		GLOBE	L01AS42	8	10	GLOBE 150# RF SS A351 CF8M TRIM 16 OS&Y GO	0	0		
210											
210	U04						***** CHECK VALVES *****				
220											
220	U04	VC1	CHECK	C01AS50	0.75	1.5	CHECK PISTON 150# RF SS A182 F316 TRIM 16	0	0		
230	U04		CHECK	C01AS45	2	6	CHECK DUAL WAFER LUG 150# RF SS A351 CF8M TRIM 16 THROU BOLTED	0	0		
240	U04		CHECK	C01AS52	8	16	CHECK DUAL DBL FLG 150# RF SS A351 CF8M TRIM 16	0	0		
250											
250	U04						***** PIPE *****				
260											
260	U04		PIPE	5365117_VC	1	16	PIPE SCH 10S SMLS SS A312 TP316/316L PE (GROOVING NOT BY SUPPLIER - TO BE DONE DURING FABRICATION)	0	0		
270											
270	U04						***** TRUNNION PIPE SUPPORTS *****				
280											
280	U04						*****FITTINGS*****				
290											
290	U04		ELBOW	5491293	0.75	0.75	ELL 90 DEG 3000# SCRD FNPT SS A182 F316/316L	0	0		
300											
300	U04	NR1					***** FITTINGS - VICTAULIC *****				
310											
310	U04		ELBOW	5485111	1	16	ELL 90 DEG GRV SCH 10S 316 SS A351 CF8M	0	0		
320	U04		ELBOW	5485112	1	16	ELL 45 DEG GRV SCH 10S 316 SS A351 CF8M	0	0		
330											
330	U04		TEE	5485113	1	16	TEE GRV SCH 10S 316 SS A351 CF8M	0	0		
340	U04		TEE RED	5485134	1.5	1.5	TEE RED GRV SCH 10S 316 SS A351 CF8M WITH FNPT BRANCH	14	14		
350											
350	U04		CAP	5485119	1	16	CAP GRV SCH 10S 316 SS A351 CF8M	0	0		
360											
360	U04		RED CONC	5485115	1	14	REDUCER CONC GRV SCH 10S 316 SS A351 CF8M	1.5	16		
370	U04		RED ECC	5485118	1	14	REDUCER ECC GRV SCH 10S 316 SS A351 CF8M	1.5	16		
380											
380	U04						***** COUPLINGS, UNIONS & PLUGS *****				
390											
390	U04		FITTING SCRD	5498221	0.5	1.5	BUSHING 3000# SCRD (MNPT/FNPT) HEX HEAD 316 SS	0.75	2		
400											
400	U04	UD2	FITTING SCRD	5498383	0.5	0.75	PLUG ROUND HEAD SCRD SS A182 F316/316L	0	0		
410											
410	U04						***** VICTAULIC COUPLINGS *****				
420											
420	U04		Coupling	5485021	1	16	COUPLING RIGID GRV 316SS EPDM GRADE E	0	0		
430											
430			SADDLE	5485138	0.5	2	VICTAULIC BOLTED BRANCH OUTLET FNPT SS316 A351 CF8M	2	8		
440	U04		SADDLE	5485141	2	2	VICTAULIC MECHANICAL TEE GRV SS316 A351 CF8M	10	10		
450											
450	U04						***** FLANGES *****				
460											

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		1" - 16" - VICTAULIC GROOVE & COUPLING SEAL WELDING - NONE PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	DESIGN: 15.8 BARG CODE - ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: 316/316L STAINLESS STEEL VICTAULIC COUPLINGS FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 110 °C	12/16/2024	BY OZAC	U04	Page 1 of 2	150VS01

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460			FLANGE	5600657	0.5	1	FLG SCRD (FNPT) 150# RF SS A182 F316/316L	0	0	
470	U04		FLANGE	5600658	0.5	0.75	FLG SCRD (FNPT) 300# RF SS A182 F316/316L	0	0	
480	U04		BLIND	5616520	0.75	16	FLG BLIND 150# RF SS A182 F316/316L	0	0	
490			BLIND	5616529	0.75	16	FLG BLIND 300# RF SS A182 F316/316L	0	0	
500										
500	U04		FLG ADAPTER	5851200	1.5	1.5	FLANGE ADAPTER 150# RF 316 SS X 12 MM TUBE	0	0	
510	U04		FLG ADAPTER	5851203	2	2	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE	0	0	
520										
520	U04						***** FLANGES - VICTAULIC *****			
530										
530	U04		FLANGE	5754196	1	16	FLANGE ADAPTER NIPPLE 150# RF SCH 10S/GRV A182 316 SS	0	0	
540	U04		FLANGE	5754197	1.5	16	FLANGE ADAPTER NIPPLE 300# RF SCH 10S/GRV A182 316 SS	0	0	
550										
550	U04						***** PIPE NIPPLES - VICTAULIC *****			
560										
560	U04		NIPPLE	5540198	0.5	0.75	NIPPLE SCH 40S 316 SS TBE 3" LG (MNPT)	0	0	
570	U04		NIPPLE	5540194	0.5	0.5	NIPPLE SCH 40S 316 SS TBE 4" LG (MNPT)	0	0	
580	U04		NIPPLE	5540199	0.75	0.75	NIPPLE SCH 40S 316 SS TBE 7" LG (MNPT)	0	0	
590	U04		NIPPLE	5541207	1.5	1.5	NIPPLE SCH 40S A312 TP316/316L GRV /TOE (MNPT) 4" LG	0	0	
600										
600	U04						***** GASKETS *****			
610										
610	U04		GASKET	5673043	0.5	16	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
620	U04	FS1	GASKET	5673033	0.5	16	GASKET 300# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
630	U04	GA6	GASKET	5673017	2	3	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.5	0	0	
640	U04	UG3	GASKET	5664201	2	16	GASKET 150# RF FLAT RING PTFE	0	0	
650										
650	U04						***** GASKETS - VICTAULIC *****			
660										
660	U04		GASKET	NOMTO_G	1	16	GASKET FOR CLAMP FOR VICTAULIC SYSTEM - NON REPORTABLE	0	0	
670										
670	U04	B1					***** BOLTING *****			
680										
680	U04		BOLTING	5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
690	U04	UG4	BOLTING	5675768	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) W/ 2 NUTS + LTCS WASHERS	0	0	
700										
700	U04						*****BLINDS AND SPACERS*****			
710										
710	U04		SPEC BLIND	5750421	2	14	SPECTACLE BLIND 150# RF SS A240 GR 316	0	0	
720										
720	U04		PADDLE	5750521	16	16	PADDLE BLANK 150# RF SS A240 GR 316	0	0	
730	U04		PADDLE	5750621	16	16	PADDLE SPACER 150# RF SS A240 GR 316	0	0	
740										
740	U04		SPEC BLIND	5750422	2	10	SPECTACLE BLIND 300# RF SS A240 GR 316	0	0	
750										
760	U04						***** RPAD FOR SUPPORTS *****			
770	U04						***** MISCELLANEOUS *****			
780										
780	U04		GASKET	5707599			LUBRICANT FOR GASKET PUSH-ON CI & DI			

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